JavaScript Code - script.js

```
/* ===== Student Attendance Tracker ===== */
const STATUSES = ["Present", "Absent", "Late", "On Duty"];
const ATTENDED = ["Present", "Late", "On Duty"]; // statuses counted as attended

let students = JSON.parse(localStorage.getItem("sat_students")) || [];
// attendance = { "2026-10-27": { "BAI101": "Present", ... } } -> one entry per student per date
let attendance = JSON.parse(localStorage.getItem("sat_attendance")) || {};
let threshold = Number(localStorage.getItem("sat_threshold")) || 75;
let editingId = null;
let draft = {}; // unsaved statuses on Mark Attendance page

/* ---------- helpers ---------- */
const $ = (id) => document.getElementById(id);

function esc(str) {
  return String(str).replace(/[&<>"']/g, (c) => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", '"': "&quot;",
     "'": "&#39;" }[c]));
}

function formatDate(d) {
  return new Date(d + "T00:00:00").toLocaleDateString("en-GB", { day: "2-digit", month: "short", year:
    "numeric" });
}

function latestDate() {
  const dates = Object.keys(attendance).filter((d) => Object.keys(attendance[d]).length);
  return dates.sort().pop() || null;
}

function todayStr() {
  const n = new Date();
  return n.getFullYear() + "-" + String(n.getMonth() + 1).padStart(2, "0") + "-" +
    String(n.getDate()).padStart(2, "0");
}

let toastTimer;
function toast(msg) {
  const t = $("toast");
  t.textContent = msg;
  t.hidden = false;
  clearTimeout(toastTimer);
  toastTimer = setTimeout(() => (t.hidden = true), 2500);
}

function saveData() {
  localStorage.setItem("sat_students", JSON.stringify(students));
  localStorage.setItem("sat_attendance", JSON.stringify(attendance));
  localStorage.setItem("sat_threshold", threshold);
}

function statsFor(id) {
  let total = 0, attended = 0;
  Object.keys(attendance).forEach((date) => {
    const s = attendance[date][id];
    if (s) { total++; if (ATTENDED.includes(s)) attended++; }
  });
  const percent = total ? Math.round((attended / total) * 100) : 0;
  return { total, attended, percent };
}

/* ---------- rendering ---------- */
function renderSummary() {
  const date = latestDate();
  const day = date ? attendance[date] : {};
  const present = students.filter((s) => ATTENDED.includes(day[s.id])).length;
  const absent = students.filter((s) => day[s.id] === "Absent").length;
  let total = 0, attended = 0;
  students.forEach((s) => { const st = statsFor(s.id); total += st.total; attended += st.attended; });
  const overall = total ? Math.round((attended / total) * 100) : 0;
  const label = date ? "on " + formatDate(date) : "no records yet";

  $("summary").innerHTML = `
    <div class="card"><span>Total Students</span><strong>${students.length}</strong></div>
    <div class="card green"><span>Present
      Students</span><strong>${present}</strong><br><small>${label}</small></div>
    <div class="card red"><span>Absent
      Students</span><strong>${absent}</strong><br><small>${label}</small></div>
    <div class="card blue"><span>Overall Attendance</span><strong>${overall}%</strong><br><small>all
      students, all dates</small></div>`;
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}

function renderDepartments() {
  const depts = [...new Set(students.map((s) => s.dept))].sort();
  const current = $("deptFilter").value;
  const currentAtt = $("attDept").value;
  const opts = '<option value="">All Departments</option>' +
    depts.map((d) => `<option value="${esc(d)}">${esc(d)}</option>`).join("");
  $("deptFilter").innerHTML = opts;
  $("attDept").innerHTML = opts;
  $("deptFilter").value = depts.includes(current) ? current : "";
  $("attDept").value = depts.includes(currentAtt) ? currentAtt : "";
  $("deptList").innerHTML = depts.map((d) => `<option value="${esc(d)}">`).join("");
}

function renderStudents() {
  const q = $("searchInput").value.trim().toLowerCase();
  const dept = $("deptFilter").value;
  const status = $("statusFilter").value;
  const date = latestDate();

  const rows = students.filter((s) => {
    if (q && !(s.name.toLowerCase().includes(q) || s.id.toLowerCase().includes(q))) return false;
    if (dept && s.dept !== dept) return false;
    if (status === "low") return statsFor(s.id).percent < threshold && statsFor(s.id).total > 0;
    if (status) return date && attendance[date][s.id] === status;
    return true;
  });

  $("studentBody").innerHTML = rows.map((s) => {
    const st = statsFor(s.id);
    const low = st.total > 0 && st.percent < threshold;
    return `<tr class="${low ? "low" : ""}">
      <td>${esc(s.id)}</td><td>${esc(s.name)}</td><td>${esc(s.dept)}</td>
      <td>${st.attended} / ${st.total}</td><td>${st.percent}%</td>
      <td><span class="badge ${low ? "low" : "ok"}">${low ? "Low Attendance" : "Good"}</span></td>
      <td>
        <button class="btn small" onclick="showHistory('${esc(s.id)}')">History</button>
        <button class="btn small" onclick="openStudentModal('${esc(s.id)}')">Edit</button>
        <button class="btn small danger" onclick="deleteStudent('${esc(s.id)}')">Delete</button>
      </td></tr>`;
  }).join("");
  $("emptyMsg").hidden = rows.length > 0;
}

function renderAttendance() {
  const dept = $("attDept").value;
  const list = students.filter((s) => !dept || s.dept === dept);
  $("attBody").innerHTML = list.map((s) => `
    <tr>
      <td>${esc(s.id)}</td><td>${esc(s.name)}</td><td>${esc(s.dept)}</td>
      <td><div class="status-group">
        ${STATUSES.map((st) => `<button class="${draft[s.id] === st ? "sel-" + st.replace(" ", "") : ""}"
          onclick="setStatus('${esc(s.id)}','${st}')">${st}</button>`).join("")}
      </div></td>
    </tr>`).join("") || '<tr><td colspan="4">No students found.</td></tr>';
}

function refreshAll() {
  renderDepartments();
  renderSummary();
  renderStudents();
  renderAttendance();
}

/* ---------- students: add / edit / delete ---------- */
function openStudentModal(id) {
  editingId = id || null;
  $("stuError").textContent = "";
  if (id) {
    const s = students.find((x) => x.id === id);
    $("studentModalTitle").textContent = "Edit Student";
    $("stuId").value = s.id;
    $("stuId").disabled = true; // ID cannot change, keeps attendance links intact
    $("stuName").value = s.name;
    $("stuDept").value = s.dept;
  } else {
    $("studentModalTitle").textContent = "Add Student";
    $("stuId").disabled = false;
    $("stuId").value = $("stuName").value = $("stuDept").value = "";
  }
  $("studentModal").hidden = false;
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}

function saveStudent() {
  const id = $("stuId").value.trim();
  const name = $("stuName").value.trim();
  const dept = $("stuDept").value.trim();
  if (!id || !name || !dept) { $("stuError").textContent = "All fields are required."; return; }
  if (!editingId && students.some((s) => s.id.toLowerCase() === id.toLowerCase())) {
    $("stuError").textContent = "Student ID already exists."; return;
  }
  if (editingId) {
    const s = students.find((x) => x.id === editingId);
    s.name = name; s.dept = dept;
    toast("Student updated");
  } else {
    students.push({ id, name, dept });
    toast("Student added");
  }
  saveData();
  $("studentModal").hidden = true;
  refreshAll();
}

function deleteStudent(id) {
  if (!confirm("Delete this student and all attendance records?")) return;
  students = students.filter((s) => s.id !== id);
  Object.keys(attendance).forEach((d) => delete attendance[d][id]);
  saveData();
  refreshAll();
  toast("Student deleted");
}

/* ---------- mark attendance ---------- */
function setStatus(id, status) {
  draft[id] = status;
  renderAttendance();
}

function markAllPresent() {
  const dept = $("attDept").value;
  students.filter((s) => !dept || s.dept === dept).forEach((s) => (draft[s.id] = "Present"));
  renderAttendance();
}

function loadDraft() {
  const date = $("attDate").value;
  const existing = attendance[date];
  draft = existing ? { ...existing } : {};

  const banner = $("attBanner");
  if (existing) {
    banner.textContent = "Attendance for " + formatDate(date) + " is already recorded. Saving will update
       the existing entries - no duplicates are created.";
    banner.hidden = false;
  } else {
    banner.hidden = true;
  }
  renderAttendance();
}

function saveAttendance() {
  const date = $("attDate").value;
  if (!date) { toast("Please select a date"); return; }
  if (!Object.keys(draft).length) { toast("Mark at least one student"); return; }
  const already = !!attendance[date];
  attendance[date] = { ...(attendance[date] || {}), ...draft }; // one status per student per date
  saveData();
  refreshAll();
  loadDraft();
  toast(already ? "Existing attendance updated (no duplicates)" : "Attendance saved");
}

/* ---------- history ---------- */
function showHistory(id) {
  const s = students.find((x) => x.id === id);
  const st = statsFor(id);
  $("historyTitle").textContent = s.name + " (" + s.id + ")";
  $("historyStats").innerHTML = `<span>Total days: ${st.total}</span><span>Attended:
    ${st.attended}</span><span>Absent: ${st.total - st.attended}</span><span>Attendance:
    ${st.percent}%</span>`;
  const dates = Object.keys(attendance).filter((d) => attendance[d][id]).sort().reverse();
  $("historyBody").innerHTML = dates.map((d) => {
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    const status = attendance[d][id];
    const day = new Date(d + "T00:00:00").toLocaleDateString("en-GB", { weekday: "long" });
    return `<tr><td>${formatDate(d)}</td><td>${day}</td><td><span class="badge ${status.replace(" ",
      "")}">${status}</span></td></tr>`;
  }).join("") || '<tr><td colspan="3">No records yet.</td></tr>';
  $("historyModal").hidden = false;
}

/* ---------- events ---------- */
document.querySelectorAll(".nav-btn").forEach((btn) => {
  btn.addEventListener("click", () => {
    document.querySelectorAll(".nav-btn").forEach((b) => b.classList.remove("active"));
    btn.classList.add("active");
    $("view-students").hidden = btn.dataset.view !== "students";
    $("view-attendance").hidden = btn.dataset.view !== "attendance";
    if (btn.dataset.view === "attendance") loadDraft();
  });
});

$("searchInput").addEventListener("input", renderStudents);
$("deptFilter").addEventListener("change", renderStudents);
$("statusFilter").addEventListener("change", renderStudents);
$("thresholdInput").addEventListener("input", () => {
  const v = Number($("thresholdInput").value);
  if (v >= 1 && v <= 100) { threshold = v; saveData(); renderStudents(); }
});
$("addStudentBtn").addEventListener("click", () => openStudentModal());
$("cancelStudent").addEventListener("click", () => ($("studentModal").hidden = true));
$("saveStudent").addEventListener("click", saveStudent);
$("closeHistory").addEventListener("click", () => ($("historyModal").hidden = true));
$("attDate").addEventListener("change", loadDraft);
$("attDept").addEventListener("change", renderAttendance);
$("markAllBtn").addEventListener("click", markAllPresent);
$("saveAttBtn").addEventListener("click", saveAttendance);

/* ---------- init ---------- */
$("thresholdInput").value = threshold;
$("attDate").value = todayStr();
refreshAll();
loadDraft();
```